Final project 3: Bug Trap Algorithm
(100 pts)

Please form a team and select a project. The maximum number of students is 2 in a team, and you also can work on your own. Once you select a project, please create a github repo and share it with me. The final grade depends on each student's performance. Please list your contribution and percentages in a separate PDF for the submission.

Question: We have a group of robots (e.g., 20). They are searching for minerals in a large unknown arena. They mimic the foraging behavior of ants. They have a limited vision, sensing, and memory. There are some obstacles in the environment. They must avoid them and plan a path to the base.

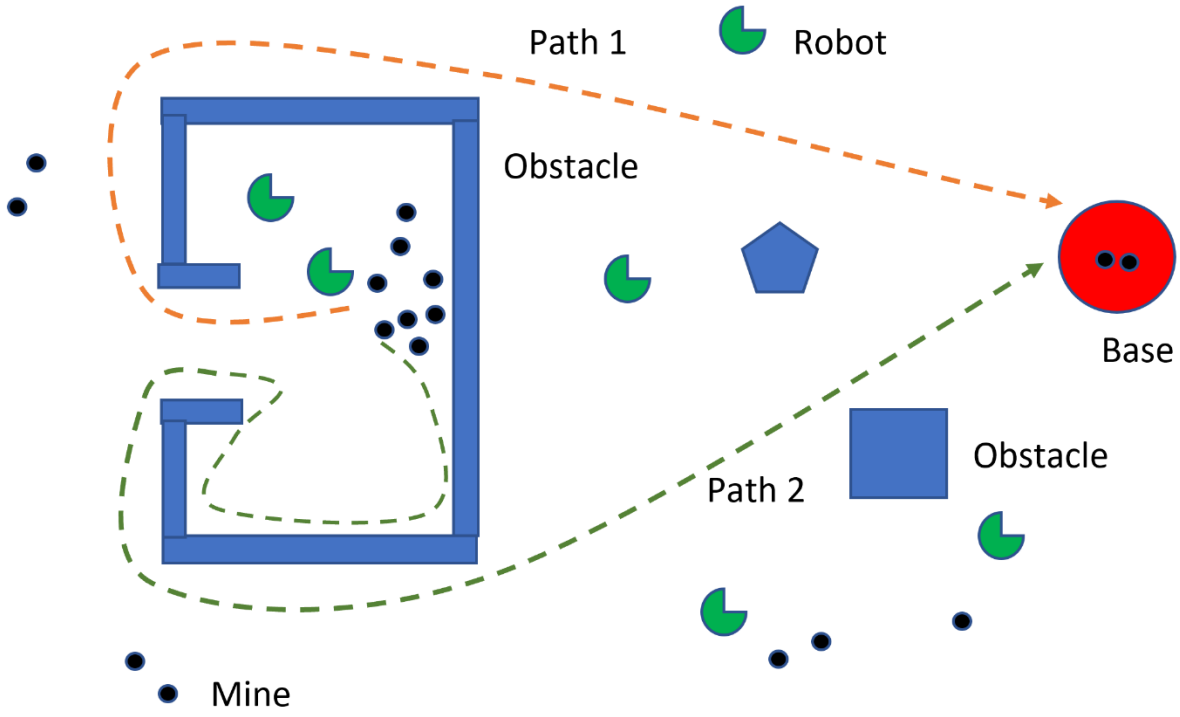
Assumptions: We assume a robot can move in all 8 directions (or 8 cells) and can move to a cell in each step. The arena size is 50 by 50 meter, and each cell is 1 by 1 meter. They can only detect objects in the 8 cells around it. We assume a robot can remember 50 locations it visited in a trip. Once it returns to the base, it shares the locations with the server in the base. Then, it deletes the locations in its memory. All robots can get the shared location information on the server. Robots can not communicate with other robots.

Goal: In the following figure, can you design an algorithm to plan a path (e.g., Path 1 or Path 2) from a mine location to the base? The algorithm should be designed based on the capabilities of robots. It should work for avoiding rectangular, triangular, or bug trap shapes.

Simulation platforms: you can implement it either in ARGoS or Webots. For ARGoS, you can find implement it based on the CPFA, DDSA, or other ARGoS examples. For Webots, you can work on the e-puck2 robot. Please create a github repository and share it with me.

Note: Robots do not have the map of the environment. They only have the local information based on the interaction with the environment. However, you can use multiple robots to plan the path or share the information via the base.

Evaluation: For this project, we will consider the difficulty, and your grade depends on your efforts.


Path 1
Robot
Obstacle
Base
Path 2
Obstacle
Mine